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Analytics

**Gap Between The Donor Support And Sustainable Development Goals For Artisanal
Small-Scale Mining Industry In Sub-Saharan Africa**

by

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EXECUTIVE SUMMARY

In many regions of the globe, artisanal small scale mining (ASM) is at least as important as large-scale mining in terms of the number of people employed. It has an important role in reducing poverty, rural development, and giving employment to nearly 40 million people living in underprivileged regions in low-income nations around the developing globe (World Bank, 2019). Providing the world with essential raw materials for the production of luxury goods and other critical metals required for the transition to a lower carbon economy (World Bank, 2017; 2020) requires the ASM industry to employ all available labour sources, including women, men, girls, and boys (Hilson, 2009; Cartier & Burge, 2011; Hilson & McQuilken, 2014; Franks, Pakoun & Ngongze, 2016; Stylo, de Haan & Davis, 2020). Because ASM is such a significant industry, it stands to reason that it intersects with the United Nations' SDGs. When large-scale mining operations include foreign investment, the Sustainable Development Goals (SDGs) may be met through promoting economic development and poverty eradication in developing countries. This is due to the high capital requirements of the sector. However, the small-scale mining industry has not delivered especially impressive results. This is owing to the enclave-like structure of large-scale resource extraction, the difficulties of catalysing links and downstream/upstream development, and the fact that corporations get significant tax savings while not being obligated to participate in the local community (Hilson and Maconachie, 2020).

According to the Atlas, industrial-scale mining: "has also contributed to a considerable number of the challenges that the Sustainable Development Goals (SDGs) intend to solve. Despite this, Atlas makes considerable and, at times, exaggerated attempts to showcase the overall economic value of the sector's activities. The research discusses how the sector employs 2.5 million people globally, how it may drive infrastructure and innovation, and how it might bring investment to rural areas and alleviate poverty there "on a game-changing scale over extended time horizons" (Hilson et al., 2018). In other words, Atlas claims that "mining and metals play an important role in the SDG agenda," which is supported by the reasoning (Maconachie and Conteh, 2021). In truth, the most serious problem is its odd choice to ignore the significance of artisanal and small-scale mining (ASM) in the context of the Sustainable Development Goals.

This paper is a direct response to this significant omission; it uses a case study from Sub-Saharan Africa to demonstrate how a stronger and better-regulated ASM industry might contribute to and accomplish the objectives of the SDGs. Host governments have "integrated the SDGs into national development agendas" (Eniowo et al., 2022), persuaded that the objectives offer the hitherto unattainable comprehensive and flexible framework necessary to handle Africa's complex issues today. Women continue to face increased rates of sexual and gender-based violence, 41% of the population now lives on less than \$1.25 per day (Hilson, 2020), one-third of children aged 12 to 14 are not in school, and more than half of those aged 15 to 17 are not in school (Hilson and Maconachie, 2020); these are just a few of the issues that require immediate attention and necessitate the conduct of this research.

The purpose of this study is to investigate how artisanal and small-scale mining (ASM) – low-tech, labor-intensive mineral processing and extraction – has developed in Sub-Saharan Africa in recent decades. The analysis focuses specifically on the types of entrepreneurs who pursue work at the region's ASM sites, as well as the innovation that occurs there, employing ideas heavily debated in the management literature, as well as complementary theories and concepts from other disciplines such as development studies, anthropology, and human geography. Based on findings from ongoing research in Sierra Leone and Liberia, the locations of two of Sub-Saharan Africa's largest and most complex ASM economies, it is argued that legal and policy frameworks implemented for the sector are not aligned with the needs and capabilities of operators, stifling efforts to formalise activities.

This research investigates a real-life scenario from the ASM industry and constructs a narrative while emphasising key points from relevant literature and providing citations in appropriate places. Researchers from the University of Surrey have compiled the dataset, which consists of secondary data, and it is now accessible via the DELVE platform. For the purposes of doing data preprocessing chores and getting the dataset ready for analysis, the programming language R was used. In order to identify correlations between the variables, several statistical tests, including Chi square and Fisher's exact test, have been used to the data. After being prepared, the dataset is then uploaded into Tableau so that the information can be graphically represented and trends and

patterns may be highlighted. In the end, the outcomes of the research are discussed, and a report is given to justify the goal of the study. In addition, ideas for future discussions have been made, and the authors have also highlighted a few new tactics that an artisanal small scale miner may take in accordance with the legislative framework that is now in place under the government.

Declaration of Originality

"I hereby declare that this thesis has been composed by myself and has not been presented or accepted in any previous application for a degree. The work, of which this is a record, has been carried out by myself unless otherwise stated and where the work is mine, it reflects personal views and values. All quotations have been distinguished by quotation marks and all sources of information have been acknowledged by means of references including those of the Internet. I agree that the University has the right to submit my work to the plagiarism detection sources for originality checks."

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15th January 2023

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CHAPTER 1: INTRODUCTION

1.1 Background

United Nations Development Programme (UNDP), Columbia University, United Nations Sustainable Development Solutions Network (SDSN), and World Economic Forum released their joint final study "Mapping Mining to Sustainable Development Goals" on July 19 (Pedersen et al., 2019). Foreign funding for large-scale mining projects is often cited as a way to advance the Sustainable Development Goals (SDGs) by stimulating economic growth and reducing poverty in developing countries. This article argues that large-scale mining activities that receive funding from outside parties contribute to the realization of the SDGs (Pedersen et al., 2019). Alas, the release of the Atlas coincided with mounting skepticism over the viability of a growth model predicated on export-oriented, capital-intensive resource exploitation (Rutherford and Buss, 2022). Some people question whether or not the mining industry has contributed much to human development.

There has been a lot of criticism towards mining industry for its contribution towards the SDGs. This is being fueled by the structure of the industry along with the difficulty of cultivating connections and upstream/downstream growth, and giving big tax breaks to businesses that do not have to work with the local people. The economic significance has been emphasized by the Atlas. For example, it mentions that this sector employs over 2.5 million people worldwide, and can help develop a nation's infrastructure by bringing foreign investment to remote areas and hence eliminating poverty. According to the arguments presented in the Atlas, this is one of the key reasons why the mining and metals sectors are important to the achievement of the SDGs. The Atlas also supports an extraction-based model of economic growth, which is another criticism of it. However, this is not Atlas's main goal. The SDGs' biggest flaw and one of the main reasons for concern is that they do not take into account the role of artisanal and small-scale mining in that framework. No other justification can be given for this choice. People who live near mineral resources are often the ones who do the labor-intensive, low-tech work that is needed to extract and process minerals. Sub-Saharan Africa is one of the poorest parts of the

world, but artisanal and small-scale mining (ASM), which is important to the world economy, was not included in the Sustainable Development Goals (SDGs).

1.2 Aims and Objectives

The primary aims are the worldwide eradication of severe poverty. Individuals care deeply that everyone on the planet is well and happy. The objective is to promote economic expansion and new employment opportunities (Newell et al., 2019). The Atlas mostly focuses on industrial mining, although it does spend some time on artisanal and small-scale mining as well. If ASM is to make a significant contribution to the Sustainable Development Goals, experts must be able to provide guidance and assessment that takes into account the whole scope of the issue (SDGs) (Pedersen et al., 2019). However, excluding ASM in discussions about the benefits that extractive industries provide to economically developing nations across the world is unfair. This has two root causes.

As the Atlas demonstrates, there is ample evidence to suggest that the sector contributes to the advancement of the SDGs, in much the same way as large-scale mining in mineral-rich areas of the world's poor. This is because the industry contributes to the progress of the SDGs in a way similar to that of large-scale mining. The second reason is that there has never been a greater need for new solutions to the issues of balancing, supporting, and stimulating the expansion of both kinds of mining industries in rising nations for economic advantage. This is one of the reasons why the significance of developing new solutions is emphasized so strongly. This dissertation addresses this gap by looking at how a stronger and better-regulated ASM sector may contribute to an impact the primary themes of the Sustainable Development Goals in the context of sub-Saharan Africa.

The practice of artisanal mining is particularly prominent in economically poor regions; hence, both the potential risks and the potential development benefits are amplified in these places. Even if it's beneficial for the economy, alternative supply chains are not always safe, well-supervised, under control, or monitored (Buss et al., 2021). Even though they have a positive effect on the economy, this is the case. Activities associated with ASM will probably have substantial impacts on the environment, the health of individuals, and society as a whole. In

addition, since the sector is mostly done clandestinely, it has the potential to be a lucrative source of revenue for criminal businesses and armed conflicts. individuals can see the progress made toward “No Poverty, Great Health and Well-Being, Decent Work and Economic Development, Life on Land, and Peace and Justice, Strong Institutions”.

1.3 Significance Of The Study

This study analyses case studies from sub-Saharan Africa to illustrate how a stronger and more regulated ASM sector may address some of the obstacles to meeting the Sustainable Development Goals (SDGs). It has come to attention that an important detail was overlooked throughout this study. Since its conception, authorities throughout sub-Saharan Africa have been lured to the Sustainable Development Goals (SDGs) because of the appealing phrase "Leaving No-One Behind." Countries that will host the SDGs "have incorporated the SDGs into policy goals" (Gatsinzi, 2020) because they see in the SDGs the illusive strong and adaptable framework needed to handle the entire nature of the difficulties Africa confronts. More than half of all teenagers aged 15–17 are out of school, and a third of all youngsters aged 12–14 are also not enrolled (Rutherford and Buss, 2022).

This makes expanding kids' access to learning opportunities a major priority. 41% of the population subsists on less than \$1.25 (USD) a day (Hilson and Maconachie, 2020). The 20 objectives and 174 related targets of the Africa-driven Agenda are designed to work in partnership with existing policies and initiatives in the area to advance progress toward the 169 Sustainable Development Goals (Pedersen et al., 2019). There's hope that the region may finally make some real headway toward its ambitions (Hilson and Hu, 2022). Social and human capital development, inclusive economic growth, peaceful communities, accountable institutions, and environmental sustainability are all taken into account by the program's numerous components. Despite this, the SDGs are aspirational because of the environment in which they must be achieved: a region of the world plagued by major development challenges such as those listed above, as well as a landscape in which only 42.8% of the community has access to electricity and manufacturing added-value activity reports for a measly 10% of GDP, with limited options for further growth in the along tertiar (Buss et al., 2021). Despite all of these obstacles, the SDGs will be accomplished. For regional governments to meet the ambitious targets outlined in the

SDGs, they will need to experiment with every economic development strategy that is now accessible. In this work, individuals argue that formalizing and expanding support for ASM is a step with a low cost but the potential for a substantial benefit.

CHAPTER 2: LITERATURE REVIEW

2.1 Variables

When looking to develop and put into action plans and strategies for the expansion of the mining industry in the future, the authors of the Atlas should be criticized for failing to emphasize how important it is to include both subsectors of the mining industry. As a result, Atlas contains errors that should be criticized. The fact that this information was left out is one of Atlas's most serious shortcomings. The omission in question is the fundamental factor contributing to the problem. Just because of that, the authors are the ones who need to take responsibility for the oversight, which is something that needs to be criticized and for which they should be held accountable. The organizational structure of the African Solidarity Movement is strikingly unlike that of large-scale resource extraction corporations operating in modern-day sub-Saharan Africa (Pein, 2022).

The Atlas suggests treating each of them independently owing to the varied contributions that each provides to the expansion and development of a kid. This is because each of them gives a unique contribution. For one activity to work well, it is necessary for the administration and supervision of the other activity to also be functioning correctly. If the objective is to achieve the highest potential cumulative impact as a result of the synergistic effects of both actions, it is impossible to carry out any activity independently of the other (Muthuri et al., 2021). The fact that governmental institutions and financial backers often ignore ASM laws is the primary factor that helps to explain why this has not yet taken happened. As a result of this pattern, the size of the informal sector has expanded in several developing nations, most visibly in the countries situated in sub-Saharan Africa.

This growth has occurred as a direct consequence of the pattern. Unfortunately, the "severe environmental, health, and social effects" that were predicted by the Atlas are very much a

reality even though lawmakers continue to refuse to hear the concerns of ASM operators. This is the case even though the Atlas accurately predicted these effects. This is because politicians continue to ignore the concerns raised by ASM operators. The reason for this is that lawmakers continue to ignore the concerns raised by ASM operators. This is the scenario that has come about as a direct consequence of lawmakers' unrelenting unwillingness to take into consideration the concerns brought forth by ASM operators. The mining industry is now headed down a path that will have "severe environmental, health, and social ramifications."

This will put office workers near industrial miners, which could result in interactions that are awkward or even violent between the two groups of people. In addition, this will put office workers near industrial miners. Sub-Saharan Africa does not considerably outperform the rest of the world in terms of the prevalence of these practices when compared to the prevalence of these practices in other areas of the globe. They have only been discovered in this one area, and this spot is the only place in any newly emerging region of the planet where they have been discovered. The only way that it will be possible to make the necessary adjustments to enable a wider engagement of a sector such as ASM in the machinery of global development strategy and to give it the "positive" prominence that it deserves is to have global frameworks such as the SDGs and accompanying guidance documents such as the Atlas.

This is the only way that it will be possible to make the necessary adjustments to enable a wider engagement of a sector such as ASM in the machinery of global development strategy. These are the only two approaches that will make it feasible to make the necessary modifications. This strategy is the only one that will be able to bring about these changes within a time frame that is considered to be fair. The authors of the MDG only gave the extractive industries a cursory assessment, there is an inadequate quantity of fundamental knowledge on which to base future studies on ASM. This is a barrier to progress in the field. The inability of the field to make advances in other areas is a direct result of this. The World Bank hosted a conference with the theme "Millennium Development Goals and Small-Scale Mining: A Conference for Forging Partnerships for Action" on June 16-17, 2005 (Pein, 2022).

The conference's official title is The Millennium Development Goals and Small-Scale Mining: A Conference for Forging Partnerships for Action. The conference's primary focus was on the issue of "Millennium Development Goals and Small-Scale Mining." This is the full name of the event

that is being held. The event was held in the nation's capital, which also serves as the location of the headquarters for the World Bank. Participants in the project came from every one of the seven geographic areas found over the whole planet, making up the entirety of the entire world.

This discussion served as the first step in the process of developing a conversation regarding the underground mining of artisanal and small-scale miners and how they link to the Millennium Development Goals (Haan et al. 2020). Possible scenarios presented by the SDGs and, later, the Atlas may have reinvigorated and advanced this interaction. As a consequence of this, the suggestions made in the essay would not be taken seriously by legislators. Extending the assessment of the current is required before delving into why ASM is vital to the lives of millions of people living in impoverished remote regions of sub-Saharan Africa.

2.2 Relationship And Variables

In the past, researchers have talked about why ASM hasn't been a priority in national development programmes and programmes to fight poverty, especially in sub-Saharan Africa, despite its clear economic value and long-standing status in many rural communities (Hirons, 2020). This part of the research starts with the literature and donor community's recognition of the social aspect of ASM around the turn of the century. This was around the same time that the idea of "livelihoods" was added to the international development agenda. During this time, there were a number of policy initiatives that tried to add the social side of ASM to the growing conversation about how to make a living. But, despite early hopes, UNDESA, the ILO, the World Bank, and the UK's Department for International Development have mostly failed to reach these two goals through their work together (Clifford, 2022).

But because these initiatives further separate the ASM sector from the bigger picture of development, they don't help spread the word about the social and economic benefits of ASM. People who pushed for a bigger focus on ASM in development policy were most likely the ones who were the least happy when things went well. In 1993, UNDESA and other groups held a meeting called "Seminar on Guidelines for Development of Small and Medium-scale Mining" in Harare, Zimbabwe. This meeting showed that the ASM industry needs a new plan for regulating

and promoting it (Maconachie and Conteh, 2021). During the seminar, there was a lot of discussion, which led to a set of guidelines that will help the industry grow and develop in the future. In May 1995, the World Bank held the International Roundtable on Artisanal Mining. At that time, experts were just starting to realize that the business caused serious problems with making a living (Rutherford and Buss, 2022).

Cardoso and Hilson, 2022, give some evidence that this is true. *Social and Labour Issues in Small-Scale Mining*, one of the most important books on ASM to date, is based on ideas that were discussed at these two seminars (Hilson et al., 2021). Even more, the authors of this important study criticize the donor community by saying that they are missing the point by not regulating the company from the bottom up and by not focusing on the social aspects of ASM. They also gave strong evidence for why ASM is so common in the black market in places like sub-Saharan Africa (Buss et al., 2021). These talks are important and shouldn't be ignored. They not only showed that ASM management was changing, but also that some investors were willing to question the tried-and-true technical focus that has helped the sector for decades.

Another important factor was that these donors were willing to put their money where their mouths were by funding innovative project work that was influenced by what was popular in livelihood development at the time. In sub-Saharan Africa, the Poverty Eradication and Sustainable Livelihoods: Focusing on Artisanal Mining Communities project was started in December 1999 by UNDESA and the United Nations Conference on Trade and Development (Hilson, 2020). This study looked at the population and organizational structure of ASM camps in four countries and made policy suggestions to improve operations and encourage people to find other ways to make a living (Mali, Ghana, Ethiopia, and Guinea). As the last part of the project, a workshop was held in Yaoundé, Cameroon, from November 19-22, 2002. Seventy experts from sub-Saharan Africa and other places took part (Hilson et al., 2018). The results of all four studies are given, as well as suggestions for how to add ASM policy to the PRSP process to help fight poverty.

The Yaoundé Vision Statement will be written using these ideas. Its main goal is to "contribute to reducing poverty and improving life in African artisanal and small-scale mining communities by 2015, in line with the Millennium Development Goals" (Eniowo et al., 2022). The Mining,

Minerals, and Sustainable Development (MMSD) Project was a big policy and research project that was funded by the nine biggest mining companies in the world from 2000 to 2002. (Pedersen et al., 2019). The Department for International Development paid for the group's main project, Livelihoods Analysis of the Artisanal and Small-Scale Mining Sector. In the first case, a lot of research done in Tanzania and Ghana, as well as information from academic and business partners, were used to find evidence that ASM works in remote parts of sub-Saharan Africa (Eniowo et al., 2022).

The ASM study was paid for by a different group, but it seemed to be a follow-up or continuation of the UNDESA/UNDP initiative and its focus on rural ways of making a living (Maconachie and Conteh, 2021). In Ghana, Mali, and Burkina Faso, for example, the Southern African Development Community made sure that ASM investigations were done quickly and thoroughly at the national level (Clifford, 2022). Third, the plan called for an ASM spokesperson to speak for the top people in the industry. This unexpected chance to build this infrastructure first came up in March 2001 at an MMSD session in London. Communities, artisanal, and small-scale mining (CASM) grew up naturally in this area (Pein, 2022). CASM was made as a way for multiple donors to work together to improve the environmental, social, and economic performance of artisanal and small-scale mining in developing countries. This is a way to fight global poverty (Clifford, 2022).

The second pillar was efforts to improve environmental and technical performance and socioeconomic development. The third pillar was better governance and formalizing the sector (Gatsinzi, 2020). By inviting businesses, governments, and funders to discuss global development on the CASM platform, it could become an important resource for ASM operators all over the world. CASM's priorities were brought together, which led to the creation of regional chapters such as CASM Africa. To sum up, at the turn of the century, it seemed like everything was in place to show how ASM helped people make a living and, more generally, how important it was for the economy. This would lead to the policy changes needed to make the sector the focus of development plans (Pedersen et al., 2019).

The Millennium Development Project and its Millennium Development Goals (MDGs) were used as a guide for a wide range of sector-specific projects. But after years of governments ignoring the sector's needs, not much would change, and the ASM industry would grow and become more established in sub-Saharan Africa's informal economy (Eniowo et al., 2022). Before saying that ASM would make big contributions to the SDGs in sub-Saharan Africa if it were properly supported, legislated, and regulated, it is important to understand why the industry is mostly informal in the region (Buss et al., 2021). This research gives a high-level look at the problems and challenges that need to be solved in order to make ASM a more important part of the regional development strategy.

2.3 Theories

It is not clear why the African Solidarity Movement was not able to make a big difference in the development plans for sub-Saharan Africa in the 1990s and early 2000s (Newell et al., 2019). Chances that the sector could have played a bigger role in helping host governments reach their goals for rural development and reducing poverty were likely affected by a number of factors that, when put together, made for a too big of a barrier. This dissertation talks about the two most likely reasons why this change didn't happen, and it does so within the framework. First of all, it's possible that UNDESA, UNDP, and DFID aren't as good as they seemed. In each "enlightened" organisation, however, the most powerful people were on the fence about spending money on research to learn more about how the sector works and, eventually, making it the focus of efforts to fight rural poverty.

Even though there was a strong minority at the time that wanted ASM to be added to the growing "livelihoods" agenda and was working on projects in this direction at international conferences, this was the case. How many powerful people there are. Also, these groups were involved in a wide range of other activities within the "space" of extractive industries, all of which could make it harder to promote ASM or even actively work against it. During the 1980s and 1990s, for example, the World Bank helped many sub-Saharan African governments carry out important mining sector reform programmes (Rutherford and Buss, 2022). The main goal of these plans was to change laws and reorganize institutions in the target countries so that they would be more likely to build large infrastructure for mineral exploration and extraction.

The ASM was a big part of many of these schemes. Millions of dollars were spent on it to strengthen the industry and make it easier for institutions to keep an eye on its activities (Pein, 2022). Studies show that after getting advice from Bank officials, governments in Ghana, Tanzania, and Burkina Faso passed large tax breaks to attract investment in large-scale mineral exploration and extraction (Rutherford and Buss, 2022). At the request of World Bank officials, several countries set up these tax breaks to encourage more exploration and use of minerals (Hirons, 2020). But efforts to make ASM legal would be slowed down if huge areas of land were given to international corporations as concessions. The World Bank gave the CASM Secretariat its headquarters and initial funding. This put the CASM Secretariat in the strange position of helping to solve problems that its parent organisation had helped to cause (Clifford, 2022).

The International Labor Organization's (ILO) support for ASM grew stronger after it put out *Social and Labor Issues in Small-Scale Mining*. Even so, the group continued to have problems with philosophical contradictions (Maconachie and Conteh, 2021). But the group's past attempts to teach the public about the health and safety problems that affect the mining industry as a whole failed (Muthuri et al., 2021). Even though the International Labor Organization (ILO) has taken a lot of steps to stop child labour, these steps may have made it harder for governments to accept and meet the requirements in *Social and Labour Problems in Small-Scale Mining* that they pay more attention to the social, cultural, and livelihood aspects of ASM. The social and work problems of small-scale mining were brought up to talk about this issue.

International Programme to End Child Labor (IPEC) says that working in agriculture is the worst kind of child labour (Muthuri et al., 2021). As a direct result, the International Labor Organization has funded a number of projects and studies that aim to make more people aware of how common child labour is in this industry in a number of sub-Saharan African countries. After many years of work, the International Labor Organization (ILO) decided that 2005's World Day Against Child Labor, called "A Load Too Heavy," should focus on the problem of child labour in the agricultural, forestry, and fishery sectors (Hirons, 2020). There were 15 delegates at the launch in Geneva. Seven of them were from countries in sub-Saharan Africa (Cardoso and Hilson, 2022).

Since the MMSD project's main goal was to show multinational corporations' "commitment" to creating programmes that build trust among governments that want foreign help and investment

in their large-scale mineral exploration and mining economies, even the number of experiments it paid for should be questioned. This helped nations trust each other so they could attract more foreign investment to expand their economies. It helps that host governments appreciate major extractive mining firms. This may increase prejudice (Hilson et al., 2019). Sub-Saharan Africans who seek massive, export-driven resource development may explain this "bias" by stating their nations require outside investment to escape economic stagnation. Sceptics claim governments support large-scale mining because they get the biggest rents, taxes, and royalties.

They say that this is why governments like large-scale mining (Cardoso and Hilson, 2022). Large-scale mining has grown in sub-Saharan Africa, which has helped multinational companies but hurt local companies. Even though the funds that host governments get aren't very big, they are still big enough that, if they are used well, they can make a big difference in reducing poverty in the area. On the other hand, there seems to be worry that African governments won't put these funds back into their own countries and that donors won't make strict rules to make sure this happens. This is a problem because it could slow down economic growth in Africa in a big way. So, the idea that ASM development was a high priority for major contributors around the turn of the century was mostly false. At the time, the price of gold was not very high. This made it possible for people in sub-Saharan Africa who wanted the gold industry to help them make a living to start their programmes.

During this time, the aggressive mergers, acquisitions, and shutdowns of global mining companies had a huge impact on the competitive environment for gold exploration and extraction in Sub-Saharan Africa (Eniowo et al., 2022). In 2002, Newmont Gold Mining of Colorado bought Normandy Mining of Adelaide, Australia. This company had a lot of holdings in Ghana (Buss et al., 2021). When AngloGold and Ashanti Goldfields merged in 2004, things changed a lot for mineral exploration and mining in Guinea, Ghana, Tanzania, and Mali (Hilson et al., 2021). Increasing production in greenfield locations like Tanzania and Mali shows that large-scale gold mine development is still ongoing in sub-Saharan Africa. Due to the low level of activity and the suspension of operations, however, those who support ASM have been given a lot of room to put their suggestions into action (Hilson et al., 2021).

As the price of gold went up gradually from 2009 to 2012, ASM, which was a key part of the region's overall plan for development, suddenly slowed down (Hilson, 2020). Because many of

the top specialists and champions of ASM from different donor organisations retired early, some of these programmes had to be scrapped quickly, and CASM was dissolved in 2010. (Pedersen et al., 2019). Due to decades of underinvestment, the ASM business in Sub-Saharan Africa is unique. Because of things that will become clear in the future, bad policies and laws have been made because people don't understand how it works well enough. It's also why most of the people who work in the industry stay hidden in the region's underground economy.

CHAPTER 3: Methodology and Analysis

3.1 Overview

Although the issues included in Mapping Mining to the Sustainable Development Goals: A Preliminary Atlas are important, a more thorough evaluation of these and related contributions requires the creation of a new atlas (UNDP, 2016a). Taking this into account, there is a significant gap between the published literature and the ground reality of the miners. As a result, this study attempts to fill this void by employing a real-world extract from the life of a small-scale miner. It starts by establishing the research philosophy and showcasing the research technique, which serves as the foundation for the analysis. Furthermore, it makes assumptions based on the extract that aid in addressing the research objectives outlined in the literature review. Further to that, it explains the process of data pre-processing and cleansing, both of which are required before producing data visualizations.

3.2 Research Philosophy

Prior to collecting any data, it is essential to comprehend the research's guiding concept and, therefore, the needed data gathering procedure. In this regard, Saunders, Lewis, and Thornhill's (2016) work has been examined. They provide numerous research philosophies that vary in their ontologies, epistemology, axiology, and methodologies of data collecting. According to these concepts, it is deduced that this research is based on the philosophy of interpretivist epistemology. This philosophy in brief, states that reality is based on people's perception and knowledge is co-created by the interactions of the researcher and the participants. It is independent of what already exists in the environment and the theories cannot be replicated to other actors in different situations.

Discussing about the research approach, there are two conflicting modes of reasoning: inductive and deductive. According to Strauss and Corbin (1998, p. 12), an inductive technique employs detailed readings of raw data in order to build concepts, themes, or a model that is based on the researcher's interpretations of raw data. This indicates that the findings are influenced by the questions posed by the researcher and are obtained directly from the study of raw data, rather than from pre-existing theories or models. Since the analysis presented in this research are based

on the raw dataset collected, connections will be made and themes established. Therefore, an inductive approach is preferred.

3.3 Research Methodology (Qualitative vs Quantitative)

Creswell, J. W. (2014) highlights that research methodology is a critical aspect of any research study as it provides a framework for the entire research process, including data collection, analysis, and interpretation. In pursuit of this study's objectives, the dataset selected has already been collected by researchers and made available on the DELVE platform. Therefore, the data collection method used is secondary data.

The preliminary investigation reveals that the questions being presented by the researcher are qualitative in character. However, there are some questions such as "Cost of Innovation," "Age," "Number of dependants," and "years in mining" that include quantitative data. The vast majority of the responses collected by the researcher are open-ended and of a raw quality. A combination of the two approaches to data analysis turns out to be beneficial in terms of detecting patterns and providing justification for this study.

3.4 Research Questions

Q1. How do donors help the small-scale mining industry to grow and contribute to the achievement of the United Nations' sustainable development goals in poverty driven regions of Africa?

Q2. What are the trends and patterns identified between the donor support and the miner's goals and objectives that result in a gap between the two in Sub-Saharan regions of Africa?

Q3. What effective business strategies can small-scale miners in Africa employ to maximize their profitability, enhance their market penetration, and guarantee their long-term sustainability?

3.5 Research Objectives

1. Investigation of the donor support available through various sources to aid and support the growth of small-scale mining industry in Africa and ultimately achieve SDG goals.
2. Identification of patterns and gaps that are caused due to difference between the objectives of donor support and the small-scale mining industry in Sub-Saharan regions of Africa.
3. Devising innovative business strategies for small scale miners to enter the formalized sector and help the industry to grow.

3.6 Hypothesis

As a result of the qualitative characteristic of the dataset, several variables are of relevance during analysis. The study assumes a fundamental grasp of statistical tests such as "Fisher's exact test" and "Chi Square test," which are used to evaluate the statistical significance between variables in a dataset. In addition, one of the limitation posed by these test is that it can calculate significance between two variables at a time. The following variables have been selected by thorough study of the dataset and the literature available within the context of this study. Therefore, this chapter has devised hypothesis for each and every relationship separately which will be accepted or rejected in the findings chapter.

3.6.1 Mining License v/s Improvements in Production practices over a year

H ₀	There is no defined relationship between mining license and improvements introduced by them in their production practices over a span of one year.
H ₁	Having a mining license will have an impact / direct relation on the improvements introduced by the miners in their production practices over a span of one year.

Table 1: Hypothesis questions for Mining License v/s Improvements in production practices over a year (Source: Author)

3.6.2 Mining License v/s Education

H ₀	No significant relationship exists between level of education and the ability to get their business licensed of an artisanal small scale miner
H ₁	A mining license is directly related to a small scale miner being educated.

Table 2: Hypothesis questions for Mining License v/s Education (Source: Author)

3.7 Data Understanding

The dataset contains 136 records of small scale artisanal miners originating primarily from the areas of Sierra Leone, Malawi, Ghana and Mali in Sub-Saharan Africa. The survey was conducted over a period of 5 months starting from 02nd March 2021 till 03rd July 2021 and collected via University of Surrey researchers included on the DELVE platform. The questionnaire, is divided into 4 parts :

Part 1 : Background

Part 2 : Licensing / Technology, General

Part 3 : Innovation & Sources of Knowledge

Part 4 : Networking & Knowledge Flows

The primary researcher of the dataset addressed a total of 52 questions, each of which have a separate function that is unrelated to the parameters of this investigation. As a result of which, it is very necessary as a researcher to compile the dataset while maintaining the goals and objectives of this study in mind at all times. To cater to this, a need for data pre-processing arises.

3.8 Data Pre-Processing

According to Vivek (2015, p. 30), data pre- processing is the fundamental conversion of raw data into a format that can be understood and analyzed easily. To ensure accurate results and remove unnecessary “noise” from the dataset, it is vital to perform some pre-processing tasks. For the purpose of this research, the following are the pre-processing tasks conducted:

3.8.1 Data Cleansing

Following the works of Vivek (2015, p.30), data cleansing is the technique of identifying faulty and incorrect data in a record set or database table. The primary purpose of the cleaning process is to find incomplete, incorrect, inconsistent, and irrelevant data and then use procedures to change or eliminate this unwanted information. These fields cause the dataset to be unnecessarily long and tend to cause confusion while performing any kind of analysis. The columns that were removed from the dataset are mentioned in Table 3.

S.No	Column Names
1	start
2	end
3	_index
4	_tags
5	_submitted_by
6	_status

7	_notes
8	_validation_status
9	_submission_time
10	_uuid
11	_id

Table 3: Column Names removed from the dataset (Source: Author)

In addition, *Age* column had responses in the form of categorical answers along with additional fields having results as 1 or 0 indicating if the respondent is a part of that particular age group or not. While this field has its use in place, it makes visualization task more difficult, and hence had to be removed.

Furthermore, there were some columns in the dataset that had empty fields. Such fields were encoded with “Not Applicable” as is the common technique while dealing with empty spaces in a dataset. The code for this can be found in Appendix 1.

3.8.2 Feature Construction

To aid the data analysis task, certain theories and concepts are adapted from vast available literature. Feature construction, in brief, is the process of creating new features or variables from the raw data in a dataset. It includes tasks, but not limited to creating new features through domain knowledge (Witten, Frank, Hall, 2016).

Some fields were created in order to accommodate the rawness of the dataset into the analysis of this research. Table 4 contains the names of these fields, and Appendix 2 contains the code for adding these features. Furthermore, the *Years* column in the mining column had data in integers ranging from 1 to 42. This column was turned into a categorical variable using the power of R programming, which means the values were classified based on ranges. The interval between the ranges was kept at 5 years in order to include maximum values while not missing any records.

S.No	Feature Construction
1	Age
2	Experience in Mining
3	Barriers 1

Table 4: Feature construction

The feature Age has been adopted from the *Age* variable, and transformed into categorical values identified as :

15-25 = “Young Adults”,

26-35 = “Early Professionals”,

36-45 = “Middle Aged Professionals”,

46-60 = “Experienced Professionals” and

60+ = “Elderly”.

Similarly, *Experience in Mining* has been adopted from *Years in Mining* and transformed into categories using bins. The function “cut” in R was used to perform this task, and bins of 5 was created. The following are the categories created according to the age group of the miner:

"0-5", "6-11", "12-16", "17-22", "23-28", "29-34", "35+"

The code for the above feature construction can be found in Appendix 2.

3.8.3 Visualization

Because of software restrictions, understanding patterns and emphasizing trends in R programming becomes challenging. As a result, industry leading data visualization software Tableau was utilized to graphically portray the information in the form of bar graphs and pie charts. The dataset was cleaned and converted into a new Excel file and was then imported into

Tableau for visualization. Tableau, as an interactive programme, enables users to visualize both quantitative and qualitative data on the tool. The variables were chosen based on their order in the questionnaire and then compared to other factors to establish trends and patterns that would be explored in the next chapter.

3.9 Analysis

3.9.1 Overview

As a recap, this study is trying to understand the gap between the artisanal small scale miners and the regulatory framework that exists in Sub-Saharan regions of Africa. After establishing the methodology for this research, the following section runs the statistical tests and tries to analyse the hypothesis as mentioned earlier. These tests have been performed in R programming; a statistical tool used to handle large dataset such as this one. The code for running chi square test and fisher's test can be found in Appendix 3.

Chi Square Test of Independance

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Figure 1: Chi Square formula (Source: ResearchGate.com, 2020)

Chi-squared test of independence is a statistical test used to detect whether two category variables have a significant relationship. It may be used to test the null hypothesis that the two variables are independent versus the alternative hypothesis that there is a relationship. χ^2 (Chi) can be calculated as indicated in Figure 1.

For the purpose of this study, the works of David Freedman (2007, p. 525) have been used. He states that large values of χ^2 imply that observed and predicted frequencies differ significantly. Small values of χ^2 indicate the opposite: observed values are close to what was predicted.

Fisher's Exact Test

$$p = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}} = \frac{(a+b)! (c+d)! (a+c)! (b+d)!}{a! b! c! d! n!}$$

Figure 2: Fisher's Exact test result (Source: ResearchGate.com, 2020)

On the contrary, fisher's exact test is used to measure the significance of the difference between the observed and anticipated frequencies in a contingency table. It is used when the sample size is small or when the predicted frequency is less than 5. According to his original publication, "The test is exact in the sense that the probability of the observed values is calculated under the null hypothesis, instead of relying on asymptotic approximations (Fisher, 1925)." Furthermore, the formula to calculate the test statistic is highlighted in Figure 2.

To apply this to the dataset selected for this study, chi-square test was run for all the relationships. Literature suggests that the standard guideline for determining if the x-squared approximation is adequate is that the Chi-square test is not acceptable when the predicted values in one of the contingency table's cells are less than 5, in which case Fisher's exact test is recommended (McCrum-Gardner 2008; Bower 2003). Therefore, to cross verify the results, Fisher's exact test is also required.

As standard practice while dealing with statistical tests it is apparent to build a graph of these variables in order present a visual representation of the relationship. This process is followed for all of the hypothesis cases within context of this research.

3.9.2 Mining License v/s Improvements in Production Practices over a year

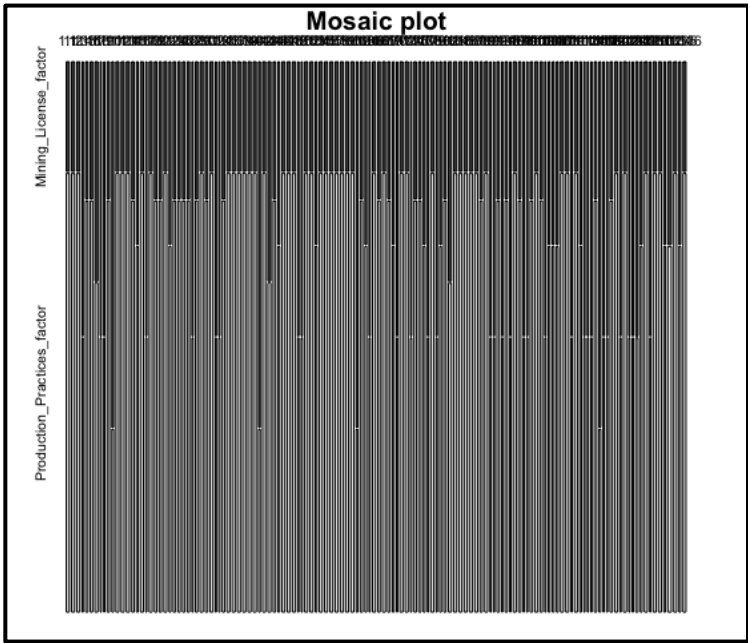


Figure 3: Mosaic Plot: Mining License v/s Improvements (Source: Author)

Pearson's Chi-squared test
data: Mining_license_vs_Improvements_table
X-squared = 3.7687, df = 3, p-value = 0.2875

Table 5: Chi Square test result (Mining License v/s Improvements)

Fisher's Exact Test for Count Data
data: Mining_license_vs_Improvements_table
p-value = 0.2149
alternative hypothesis: two.sided

Table 6: Fisher's Exact test result (Mining License v/s Improvements)

The findings of the chi-square test are summarised in Table 5. The alternative hypothesis is rejected because the probability value (p value) is greater than the 5% significance level within three degrees of freedom. To cross-validate the findings, the variables were also subjected to Fisher's exact test, and the results are reported in Table 6. In addition, Fisher's exact test findings indicate that there is a significant relationship between the two variables.

Consequently, the null hypothesis is accepted while the alternative hypothesis is rejected. In other words, there is a strong correlation between the introduction of innovative production methods and efficient mining processes and the possession of a small scale mining licence. This issue is elaborated upon in the subsequent chapters.

Drawing the attention back to the literature available, these claims can be seen in real life. Hilson et al. (2018) began applying the general entrepreneurship typology developed in the management literature to sub-Saharan Africa, and found that the "poverty-driven" and "get rich quick" narratives corresponded very closely to the necessity and opportunistic categories of activities, respectively. Those who are licenced, presumed to have security of tenure, financially supported, and deeply established in the official sector are more likely to fall into the "get rich fast" category of ASM. However, the 'poverty-driven' people in the area are the ones who fall behind. Keeping this in mind, a synchronous line can be built between the general entrepreneurial typology to the current situation in the ASM industry. Having a mining license enables the miners to have access to technical knowledge, financial support and profitable mines that help to increase their daily produce. Poverty driven unlicensed miners on the contrary miss out on these features and result in becoming a part of the current informal sector.

3.9.3 Mining License v/s Education

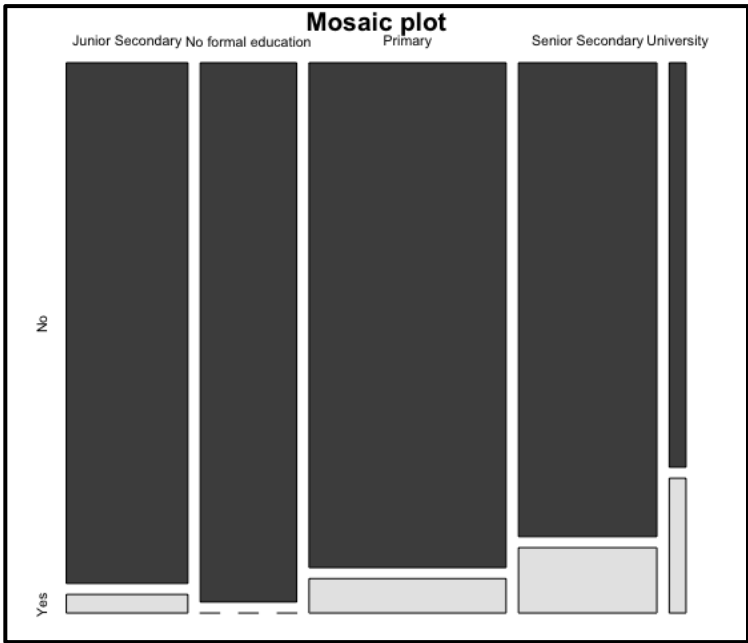


Figure 4: Mosaic Plot: Mining License v/s Education

Pearson's Chi-squared test
data: Mining_license_vs_Education_table
X-squared = 5.9102, df = 4, p-value = 0.206

Table 7: Chi Square test result (Mining License v/s Education)

Fisher's Exact Test for Count Data
data: Mining_license_vs_Education_table
p-value = 0.17
alternative hypothesis: two.sided

Table 8: Fisher's exact test results (Mining License v/s Education)

As it can be seen from Table 7, the p value is greater than the 5% significance level with four degrees of freedom. Referring to the literature highlighted before, this indicates that there is a strong relation between the level of education and availability of a mining license.

To verify this, fisher's exact test was run and the results are tabulated in Table 8. Furthermore, after running fisher's exact test for significance, the results are cross verified. Therefore, with respect to the hypothesis stated in Table 2, it is apparent that the null hypothesis should be rejected. In other words, it can be said that a small scale miner's ability and knowledge to get their business formalized is strongly dependent on the level of education that they have undergone.

Much of the literature available supports these claims. For example, in Zimbabwe, small-scale mining is regulated by the Mines and Minerals Act which, among other things, sets out the process for obtaining a license for small-scale mining. According to this act, an applicant must have at least a grade 7 education and must pass a test on mining regulations and safety (Mines and Minerals Act, 1961, ch 21:05).

CHAPTER 4: FINDINGS

4.1 Overview

The following section performs an analysis of the dataset, confirms the hypotheses, and reports the findings from both an academic and an applied point of view. This is done after the preceding section has established the theory and methods of our proposed application in great detail. To summarize, this research bridges the gap that was highlighted by (Yeboah, 2016) and others (e.g., Aryee et al., 2003; Geenen, 2012) that stated the need for actual data from the miner's perspective to understand the reason for an increase in illegal mining. In addition, the research analyses case studies from sub-Saharan Africa to highlight how a stronger and more regulated ASM industry may help overcome some of the hurdles to achieving the Sustainable Development Goals (SDGs).

This chapter starts with an analysis of the distribution of licenced vs unlicensed miners and attempts to comprehend the obstacles that prevent them from becoming formalized while highlighting the benefits of having a licence for operators. The report then investigates the government's assistance, both financially and in terms of training and seminars. Lastly, this research examines trends in the innovation component of the mining operation and the level of formalization the miners currently exhibit.

4.2 Mining License

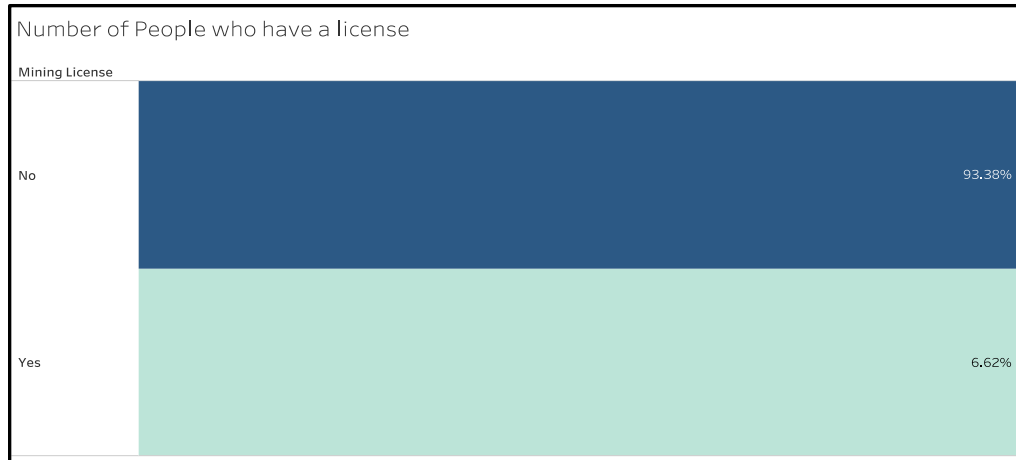


Figure 5 : Number of Miners who have a mining license (Source: Author)

As previously discussed, one of the most serious issues in the artisanal small scale mining industry is the sector's formalization. Figure 5 reveals that about 93.38% of the survey respondents did not own a mining licence. As discussed in the literature review, these findings are consistent with the ground reality in Africa which point to this sector being largely informal in nature. In addition, claims made by (Armah et al., 2013) can be confirmed as the data clearly suggests a large number of unlicensed miners currently working in the industry. Furthermore, this study bridges the gap that was highlighted by (Yeboah, 2016) and others (e.g., Aryee et al., 2003; Geenen, 2012) that stated the need for actual data from the miner's perspective to understand the reason for an increase in illegal mining.

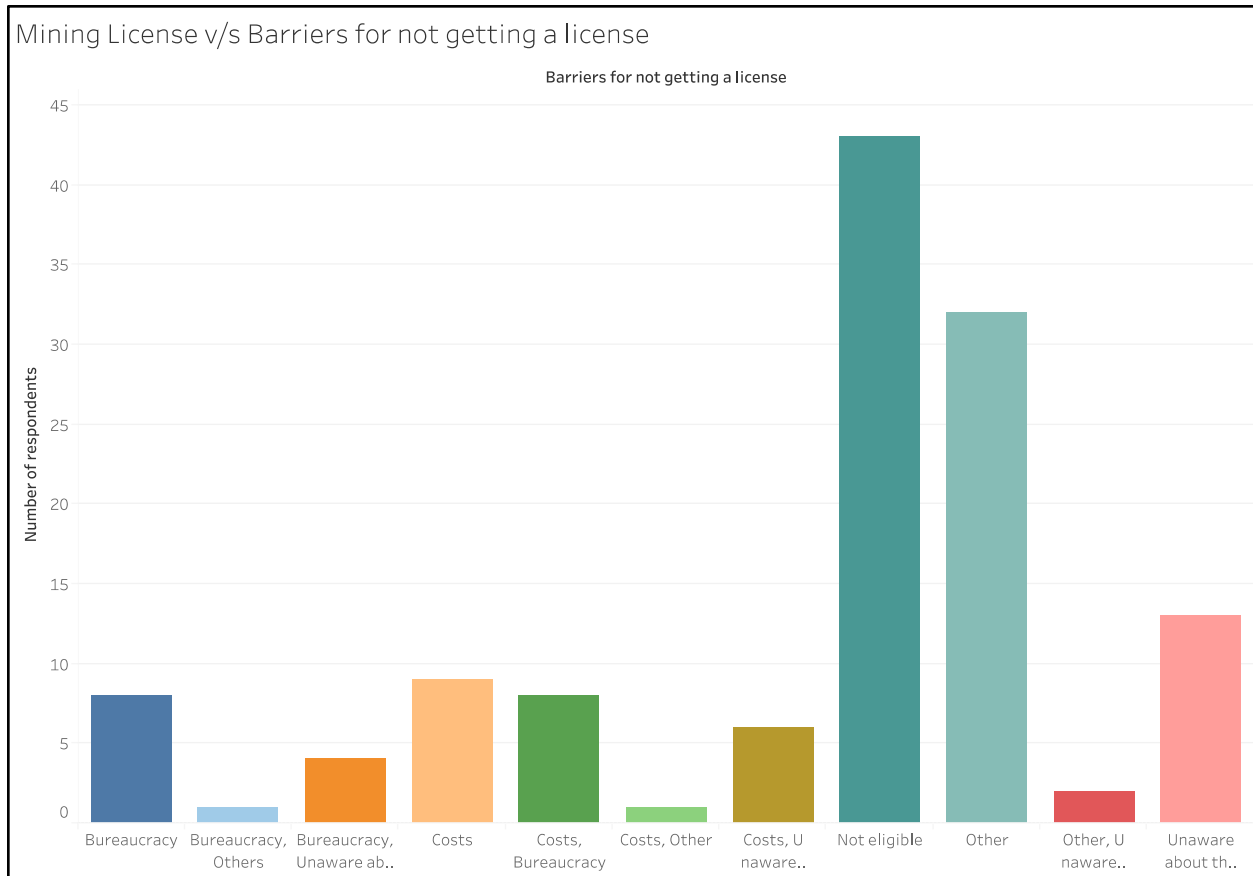


Figure 6 : Barriers for not getting a license (Source: Author)

In response to efforts made by the government to "formalize" the illegal activities of ASM operators continued to grow at a large scale. The miners provided a variety of explanations for why they did not register, including a lack of knowledge of the legislation, financial restrictions, time-consuming registration processes, and an inability to reach registration authorities (Figure 6). According to the Small-scale Gold Mining Act passed in the Republic of Ghana in 1989, ASM licences can be awarded to Ghanaian individuals, corporations, or groups of Ghanaians who are at least 18 years old. In addition, a variety of costs, including application, processing, and consideration fees, must be paid to the Minerals Commission by aspiring miners before they may be issued mining licences according to the requirements of the Act. Furthermore, there are additional fees associated with obtaining clearance from the EPA, the Office of the Administrator of Stool Lands (OASL), and district or municipal assemblies. This suggests that in addition to the prohibitive cost, which was indicated by the International Labour Organization

(ILO, 1999), the cumbersome procedures involved in obtaining a licence to operate legally and a lack of awareness account for the inability of ASM operators to register. This study confirms these claims, and also highlights that almost 11% of the population sample was uninformed of the licence requirements, and 33% were ineligible (Figure 6). Furthermore, about 7% of the population was restricted by bureaucratic procedures and the payment of bribes to the local authorities to process their document.

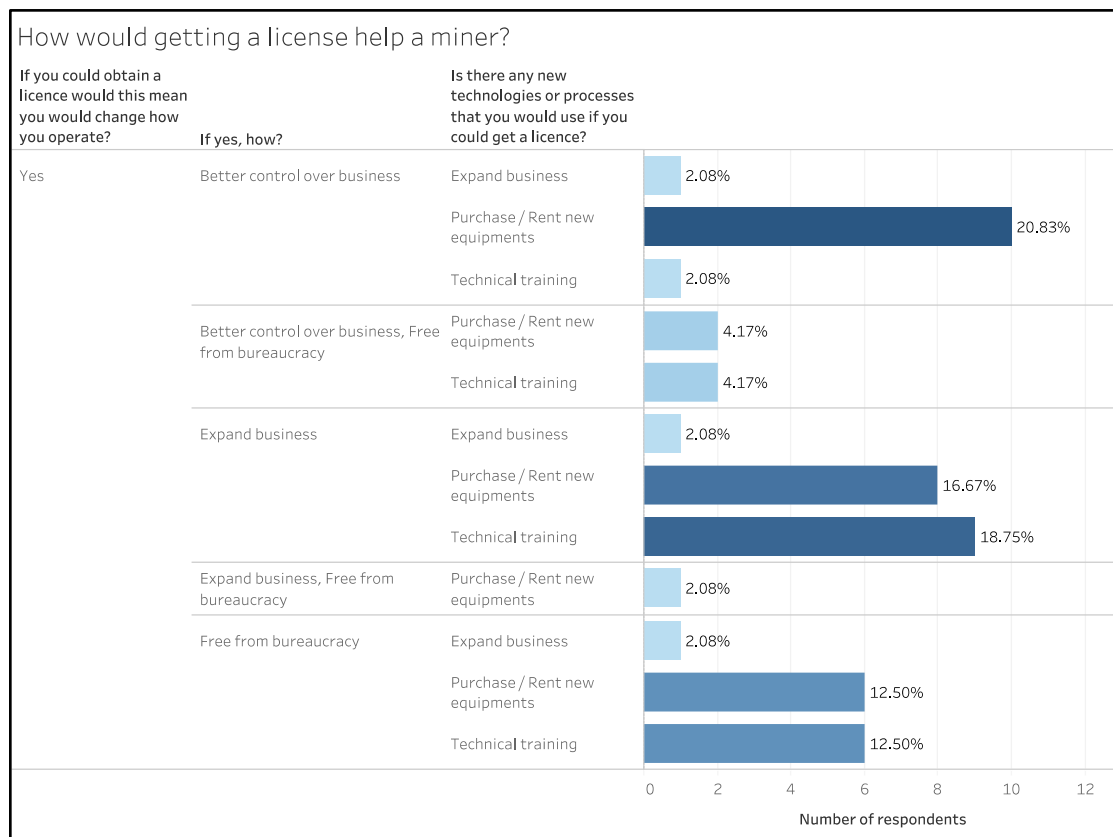


Figure 7: How would a license better help a miner (Source: Author)

As correctly highlighted in (*Government of Ghana, 2006, Sec. 82*), rural poor people typically engaged in alluvial gold mining still use rudimentary tools such as picks, chisels and pans for survival. These operators are still subject to the same mining license requirements that their counterparts (LSM) using capital-intensive sophisticated machinery. These claims can be verified by the objectives of this study. For example, Figure 7 shows that 20% of small-scale

miners expect to gain more operational control over their businesses and acquire or lease new mining equipment in the near future. At the moment, they have to spend a larger percentage of their daily yield on basic equipment, and they must also resort to unlawful acts like bribery and corruption to make ends meet. Also, one-fifth of the mining workforce appears to think they would also be given the opportunity to attend training on technical topics including safety, health, and environmental protection.

4.3 Training and Assistance from the government

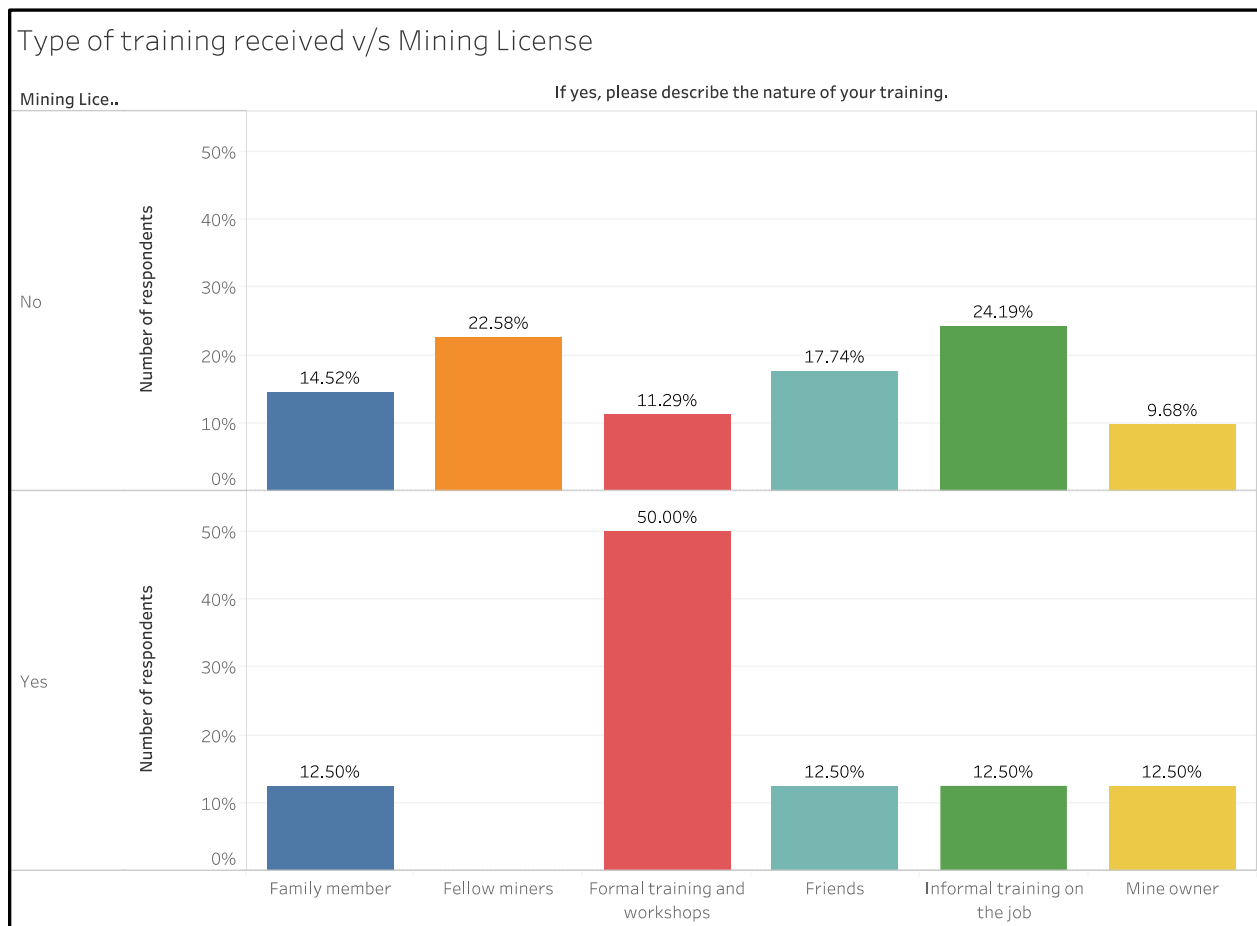


Figure 8: Type of training received v/s Mining License (Source: Author)

In an effort to address the issues presented by ASM, several nations have attempted to develop new or modify current mining laws to identify ASM and to enforce mineral and tenure rights in

the sector. All of this culminates in the state issuing prospective miners different forms of "legalization" or "paper formalization" (Kinyondo and Huggins, 2020; Salo et al., 2016). Building on this framework, Figure 8 shows that 50% of licenced miners had formal training and workshops. This helps to establish the fact that there is a direct relationship between mining license and formal training workshops, and hence confirms the hypothesis. Other forms of training include on-the-job training, friends and family members, and mine owner knowledge sharing, which account for 12.5% of the sample population. In contrast, unlicensed miners' primary source of training is on-the-job informal training. According to Figure 8, 24% of respondents learnt the techniques of mining only through informal training. Unlicensed miners do not have the opportunity to undergo formal training as they are considered as mining illegally, however they do have to produce more and pay higher amount of fees as bribes to the authorities to continue mining as highlighted earlier.

In spite of the fact that policymakers appeared to recognize the economic significance of ASM and the necessity of more effectively regulating the activities associated with it, they refrained, at this early stage, from implementing the operator-friendly formalization frameworks that are so crucial to pushing the sector toward a more sustainable path. Among the recommendations that were provided were the ideas that "To facilitate formalization, registration procedures and regulatory compliance can be simplified" (UN, 1996, p. 219), "Governments must be prepared to move beyond the establishment of legal frameworks, to identify deposits and areas amenable to small-scale development, including the preliminary evaluation of their technical and economic viability at different levels of operation," (Davidson, 1993, p. 317), and "[a] government[t] must be prepared to move beyond the establishment (Kumar and Amaratunga, 1994, p. 20). However, the majority of these suggestions were only implemented by governments in a piecemeal basis, and in other instances, they were completely disregarded.

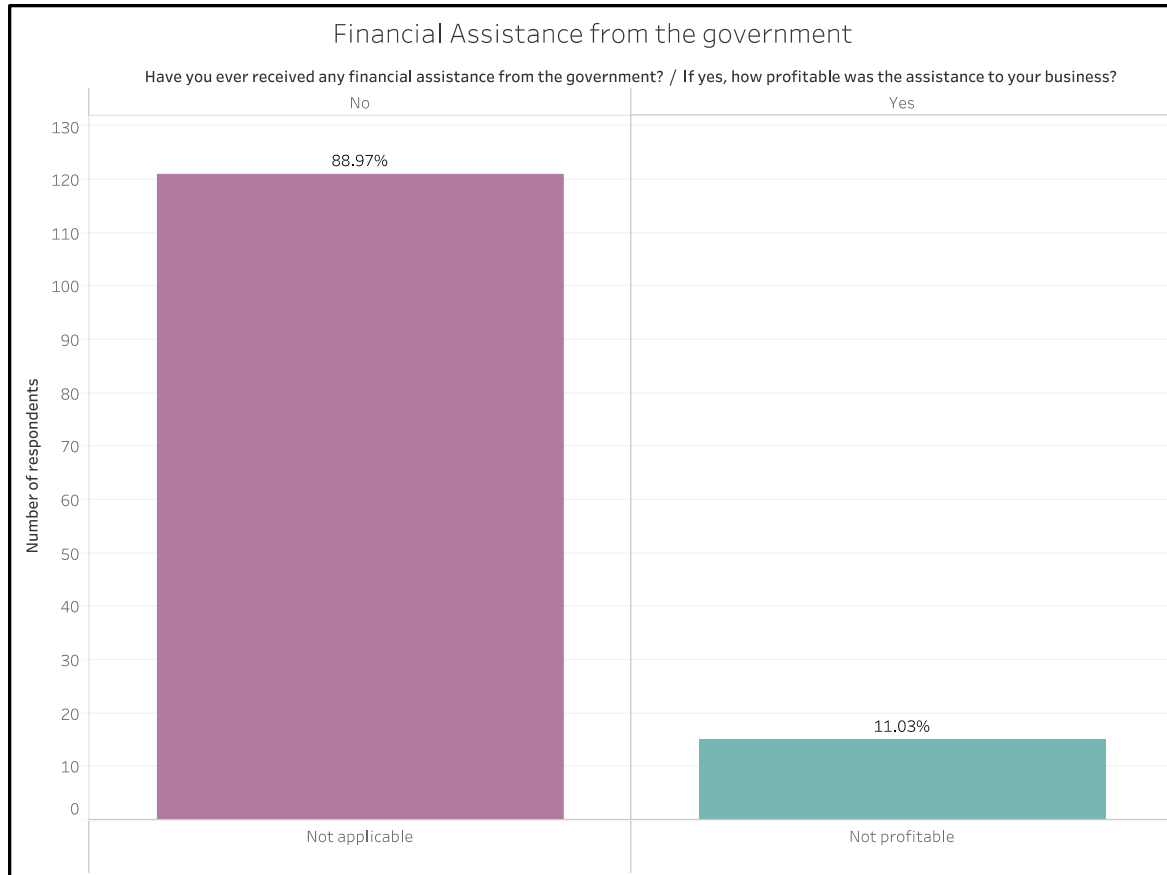


Figure 9 : Financial assistance from the government (Source: Author)

According to (Pien, 2022), the ASM was a big part of many schemes wherein millions of dollars were spent to strengthen the industry and make it easier for formal actors and institutions to keep an eye on its activities. Additionally, referring back to the previously highlighted literature, the Yaoundé Vision Statement's main goal was to contribute to reducing poverty and improving life in the ASM communities in line with the Millennium Development Goals seemed like a failure. As indicated in Figure 9, about 89% of the miners reported receiving no financial assistance from the government. In addition, 11% of those who responded affirmatively said that the assistance was not lucrative enough to improve their operations. Therefore, this reiterates the primary reason for performing this research, proving that there exists a gap in what has been said and what is the ground truth.

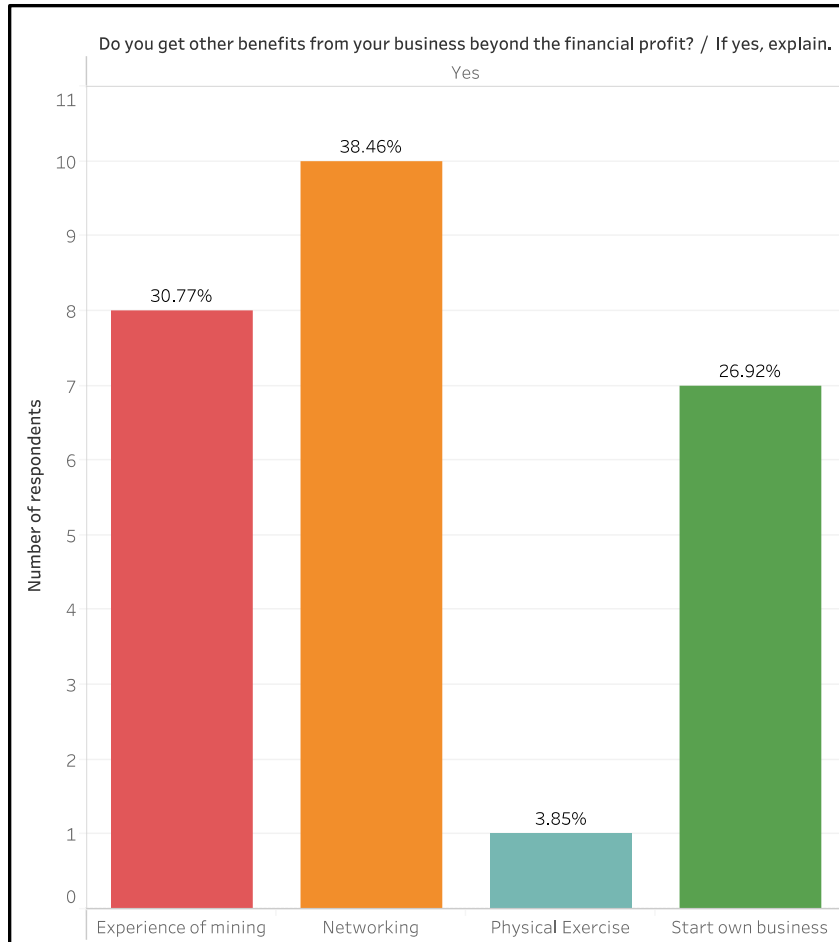


Figure 10 : Benefits beyond financial profit

As highlighted in the literature review, even though the International Labor Organization's support for the ASM sector grew stronger, in a much more general sense it failed to cater to the welfare of the small scale miners. The dataset explores into this angle and looks into the various advantages of the sector to attempt to understand them. Figure 10 shows that over 39% of respondents value the opportunity to connect with other miners, supporters, and brokers. As a result, they are able to reach a wider audience. This opens up opportunities to collaborate, which might be useful in the future for issues like securing cheaper loans outside of the conventional banking system. Additionally, 30% of responders gain mining experience, which is a driving factor for them to stay in the industry. They utilize this expertise to help other people who are interested in mining (usually their own family members) as seen in Figure 8.

4.4 Innovation and formalization

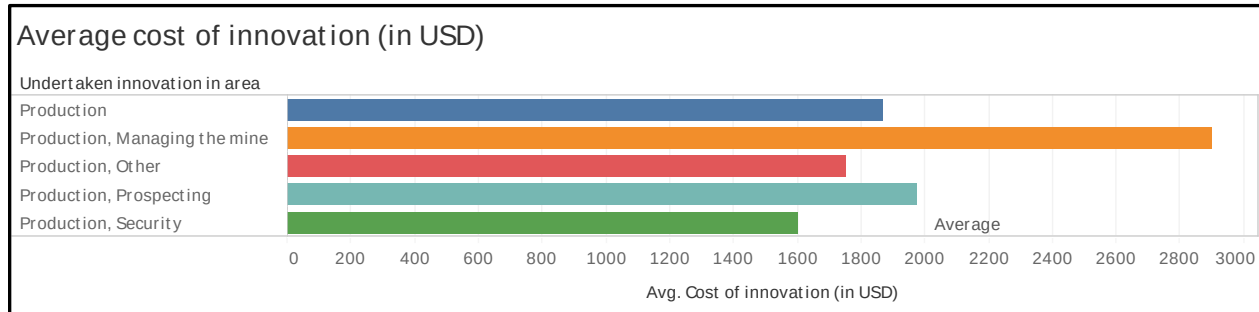


Figure 11: Average cost of innovation (in US\$)

While introducing innovation in the ASM industry, Schumpeter (1934, 1940) ideas need to be considered as a base which talk about ‘the unique connection between “entrepreneur” and innovation.’ According to (Manimala, 1992, p. 47), small scale miners should also be considered as “innovative and resilient” even though they are deeply wedded to and conditioned by their experiences in the informal economy. Furthermore, as correctly highlighted by (Hilson & Maconachie, 2020), despite missing the required permissions, titles, and paperwork to access support, many individuals presently involved in ASM throughout Sub-Saharan Africa have been able to make income and reinvest it into their operations. Also, majority of the sector’s activities that are a part of the informal economy are undertaken by unlicensed miners. This becomes a primary reason as to why, miners are restricted to innovate within the production area. This aspect is investigated further in this study by analyzing the innovation undertaken and the average cost invested (Figure 11). The average cost of innovation across all population sizes was USD 2,018 with the highest share in the "Production, Mine Management" category at USD 2,900. This category pertains to mining innovation methods such as water extraction, sand filtration, and rock crushing. Significant contributions may also be noted in the "Production, Prospecting" category at USD 1,971. Hence, these insights give weightage to points raised by (Gavin Hilson, Maconachie et al., 2020) that argue only those who are able to pay hefty fines and acquire a mining license will have access to sophisticated machinery and training workshops.

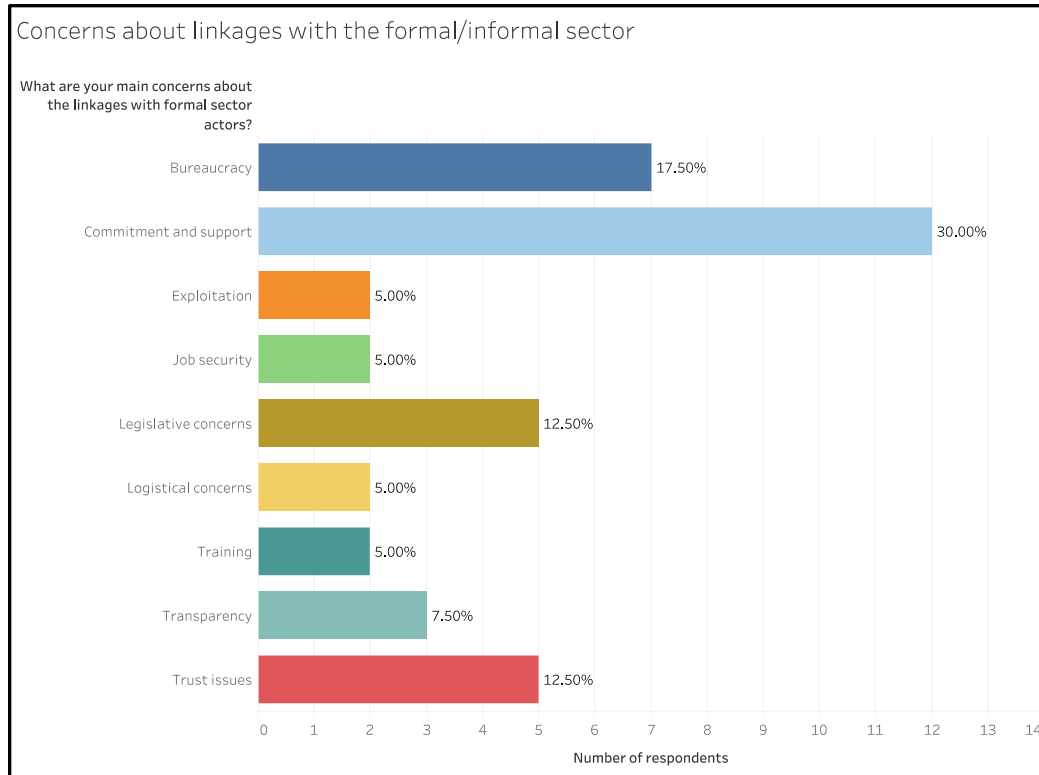


Figure 12 : Concerns about linkages with the formal sector

Much of the available literature concentrates on the political and economic elements of formalization of the sector, conveniently ignoring the core issue, which is the miners' welfare. For example, according to (Pien, 2022; Hirons, 2020), majority of the profitable land was given to international corporations as concessions, but these lands were used for the financial benefits of the companies. This results in the miners fearing that commitment and support from formal players would weaken over time (Figure 12). Mistrust and financial motivations of the actors also play a significant influence in the formation of this attitude among miners. This issue is aggravated by local officials' corruption, which is considered to persist even after formalization. In addition, 5% of miners believe that if their positions are established, there will be a threat to their job security since they are still considered "low-level miners."

Going back to the literature that was mentioned earlier as an illustration, during the years 2009 through 2012, the price of gold increased. At the same time, top specialists and champions of the ASM sector, who were previously a part of CASM, retired earlier than expected, which resulted in the dismissal of CASM in 2010. During this time period, there was a new vision of hope that

arose among the small-scale miners; but, with the crisis in 2010 the confidence over formal players completely disintegrated.

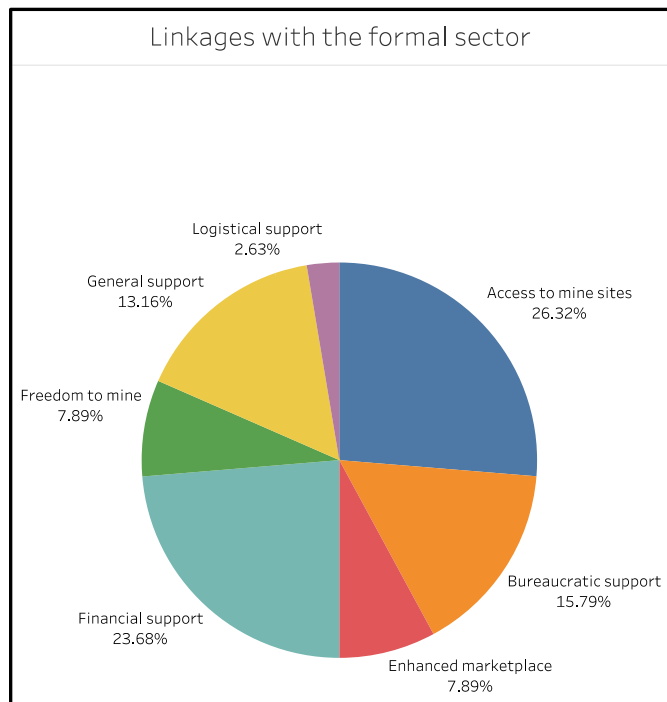


Figure 13: Linkages with the formal sector

(Hilson, 2020) in his article states that multilateral organisations like UNECA, The World Bank and donors like Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) are more focused than ever to formalize and support the ASM industry. However, they are restricted within the current development framework which was designed ignoring the importance of this industry within their regulations. These force the operators who are mining illegally to take advantage of what is available to their disposal which are a part of the informal sector. To give weightage to these claims, it can be seen in Figure 13 that 26% of the operators have access to mineral-rich mining locations, enabling them to boost their productivity. In addition, 24% also highlighted having financial support in dire circumstances at a cheaper interest rate than conventional banks. This support helps them to rent/purchase machinery to assist them in their day-to-day activities and also in some cases, act as seed money for investing in their own side business. Furthermore, 16% of the respondents stated that due to their mining practice, they were

able to make connections with local authorities that permitted them to mine freely and get bailed whenever they were in trouble.

CHAPTER 5: Discussion

5.1 Overview

The previous chapters looked into more detail what the current situation of the ASM sector is and how it continues to be excluded from the government and donor's regulatory framework that leads to informality of the industry. Restructuring the mind and highlighting the research objectives is crucial before going into further discussions. This study's primary aim is to understand the current regulatory framework existent in Sub-Saharan regions of Africa, and how ASM, being one of the top performing industry that employs over 20 million people directly (Davidson, 1993; ILO, 1999; Maconachie and Hilson, 2011), still remains one of the largest informal sector in the country, and how if properly formalized could assist in the achievement of the Sustainable Development Goals as set by The World Bank. The following section discusses these issues in further detail while highlighting important points from the literature available and discussed earlier.

5.2 Context

A final issue investigated here is the inseparable linkages ASM has with agriculture in sub-Saharan Africa (Maconachie, 2011; Hilson et al., 2016). Over the past two decades, a sizable body of analysis has emerged which explores the connections between the two activities. Much of this work was initially driven by the realization that in many countries, structural adjustment and market liberalization played a role in 'hollowing out' the state, which caused widespread unemployment and has made agriculture a challenging enterprise for poor smallholders (Weng et al., 2015). Detailed research carried out in Ghana (Okoh and Hilson, 2011; Hilson and Garforth, 2013), Tanzania (Fisher et al., 2009; Bryceson and Geenen, 2016; Aizawa, 2016), Senegal (Persaud et al., 2017), Liberia (Hilson and Van Bockstael, 2011), Mozambique (Mondlane and Shoko, 2003) and Zimbabwe (Mkodzongi and Spiegel, 2019), confirms that ASM has become an important supplementary income to farming, and a key economic activity relied upon by households, especially during periods of hardship. These findings are further supported by abundant evidence which draws attention to the inter-locking nature of the agricultural and ASM

sectors, and their tendency to ‘dovetail’ one another in many sections of sub-Saharan Africa (Maconachie and Binns, 2007; Banchirigah and Hilson, 2010; Kamlongera, 2011).

Individuals engaged in ASM and farming often ‘straddle’ both activities throughout the course of their lives, continuously striking a balance between the two. For many of the region’s poverty-stricken groups, doing so is the key to their survival, particularly when suddenly confronted with difficult circumstances (Maconachie and Hilson, 2018). When this body of research is nuanced further, however, what becomes even more clear is what impacts a movement from farming into informal ASM has had at the household level in sub-Saharan Africa. These contributions, the two most important of which will be reviewed here, further solidify the case for formalizing ASM and making it more of a point of emphasis in the region’s programs and plans built around the SDGs.

The first contribution that must be emphasized is how a ‘branching out’ into ASM has changed the fortunes of women engaged in agriculture. While sub-Saharan Africa is a mosaic of cultures, each underpinned by different beliefs and norms, a major concern voiced in donor circles about the region as a whole since the launch of the MDGs is gender inequality. A burgeoning body of analysis has since emerged (e.g. Balamoune-Lutz and McGillivray, 2015; Tibesigwa and Visser, 2016; Njoh et al., 2018) which explores gender inequality in sub-Saharan Africa in considerable depth, from which it is very clear that it is a deeply-rooted problem that persists region-wide. Reflecting on the issue generally, the Africa Human Development Report 2016 Accelerating Gender Equality and Women’s Empowerment in Africa (UNDP, 2016b) explains that ‘Because social norms and beliefs assign African women and girls the primary responsibility for care and domestic work, women, on average, spend twice as much time as men on domestic work – child and elderly care, cooking, cleaning, and fetching water and wood’ (p. 5).

Despite the common perception that women in Sub-Saharan Africa are wholly dependent on their husbands and male relatives, the evidence suggests the opposite: that in many cases, they bear far greater responsibilities than the men in their families (Falola and Amponsah, 2012). For this reason, they need to be supported in ways which are compatible with local cultures. Women’s marginalization is particularly visible in the region’s agricultural space, which has

significant implications for development. In sub-Saharan Africa, an estimated 60–70% of food is produced by women but most struggle to do so because of gender-based norms that limit their access to capital inputs (SDGC, 2017). This has proved particularly challenging for at least two-third of the region's women who engage in agriculture, 80% of whom are subsistence farmers.

In Ethiopia and Nigeria, where males own 80% of farmed land, attempts to expand agricultural plot sizes to stabilize farm productivity headed by women fail. In Malawi and Niger, where males dominate 70% and 53% of land, respectively, and in Tanzania and Uganda, where women own 24% and 38% of land, respectively, similar issues remain (Deininger et al., 2018). Sub-Saharan African women's lack of land and agricultural resources like fertilizers has impeded farm development. In the mid-2000s, cereal crop production was half of South Asia's and one-third of Latin America's due to these restrictions (Chauvin et al., 2012).

Women's simultaneous engagement in informal ASM activities, however, has helped them to overcome these constraints. The dominant narrative around women's participation in ASM activities in Sub-Saharan Africa is that they mostly populate the lowest rungs of the labor hierarchy and are prevented from ascending to some of the sector's higher-paying jobs. There is certainly evidence that confirms that this is often the case. For example, in Zambia, Tychsen et al. (2018) report that 'When it comes to the participation of women in the ASM sector, socio-cultural and physical barriers, such as the harsh conditions which miners are subjected to and the traditional and cultural beliefs that discourage women from taking the lead in economic activities, prevent women's effective participation in informal mining activities' (p. 52). Barreto et al. (2018d) report much of the same from Orisi, an artisanal gold mining locality in Kenya. Here, the authors explain, the division of labor has significant gender implications when it comes to the distribution of benefits from working in the mine: women mostly work in the lowest paid jobs (e.g. crushing), earning US\$78/monthly per month), are largely excluded from the more lucrative work in mill operations (jobs that fetch US\$1560/month), and consequently generate only 11% of the revenue share despite comprising 38% of the workforce.

In Rwanda, Buss et al. (2017) report that 'Women's participation in mining livelihoods', which is 'somewhat of a departure from the rural norm', is constrained by what the authors refer to as a 'double burden'. On the one hand, here, because ASM 'is a non-agricultural sector that is...seen

by some in the community as inappropriate for women', they are constrained from 'improving their own economic potential'. On the other hand, the 'work possibilities at ASM zones for women were also strongly shaped by pressures on them by their families, households and the wider community concerning their gendered duties'.

It is therefore clear for many women, that their participation in artisanal mining 'is shaped by a combination of productive and reproductive roles' (pp. 34–35). Barreto et al. (2018b) also explore this in the case of Rwanda: ...[In Gifurew,]... Women miners conveyed how their work in the mines enabled them to buy livestock, provide financial support to their parents, and other benefits that they saw as an improvement from their previous work in farming...[and] As agriculture and mining visibly co-exist around COMIKAGI, much of the food consumption component of poverty line determination is likely to be met through subsistence agriculture, much of which is provided through the labour of female household members, and potentially providing some (although not all) explanation for women's low participation in the sector. [p. 21, 23] But at the same time, despite earning less than men, women do, in many cases, view their move into ASM as empowering and attribute it as the main reason why they are able to sustain the agricultural cycles they drive.

Women's 'success' in generating earnings from ASM to support their agricultural ventures and other businesses, however, may be due to the maneuverability they have been afforded targeting other minerals: in Zambia, semi-precious amethyst while others target emeralds; in gold-rich Ghana, salt; and in the aforementioned Sierra Leone, as indicated, gold when most mining policies and activities revolve around alluvial diamonds.

In short, formalizing and supporting ASM is not necessarily a direct and fully-replicable means for empowering women in these, and other, environments. But it is a step in the right direction as it supports them in their endeavors as farmers, and in the area of agricultural development, represents one fairly easy way to 'empower women economically, increase food security, improve incomes and livelihoods, and contribute to Africa's broader economic growth' (ADB, 2015, p. 9).

The second contribution, though slightly more speculative though nevertheless promising, is the direction in which research that explores the linkages between farming and ASM in sub-Saharan Africa should be trending: work which investigates the sector's impact on food security. Most of the studies identified here point out how engagement in ASM has positioned families to acquire agricultural inputs such as fertilizers and pay for labourers, as well as increase the sizes of their plots. While this very clearly supplies a household with increased quantities of food and potentially more revenue, in the process, helping it to better cope with shocks and stresses, the links between ASM, agriculture and food security, and how this builds resilience in rural sub-Saharan Africa, have yet to be fully articulated. It is a region plagued by erratic weather, recurrent drought and flooding, conflict and poor soil fertility. Combined, these factors have constrained agricultural production on 33 million farms of less than 2 acres in size in sub-Saharan Africa, or 80% of the region's agricultural plots (NEPAD, 2013a,b).

During the period 2015–2016, the number of people undernourished in Sub-Saharan Africa rose from 200 to 224 million, which represented 25 percent of the 815 million people undernourished in the world at the time (FAO, 2017). Formalizing an ASM sector which is now heavily relied upon by hundreds of thousands of – predominantly poverty-driven – rural African families to support their subsistence agricultural practices speaks directly to SDG 2 ('End hunger, achieve food security and improved nutrition and promote sustainable agriculture'). A key point to make about the growing body of literature which explores the links between ASM and farming in Sub-Saharan Africa is how this phenomenon persists across a range of micro-climates: from the semi-arid stretches of Mali, through to the more tropical climates of Ghana and Tanzania. The impact of this livelihood diversification on food security, along with women's empowerment, is an area where ASM, even in an informal state, aligns closely to what the SDGs are attempting to achieve in sub-Saharan Africa and more generally. There are also other very obvious ways in which, in even a predominantly informal state, ASM contributes to the SDGs in sub-Saharan Africa, including how it engages idle youth, generates income families need to pay school fees, and how it catalyzes local infrastructural development. These impacts would only be more widely felt if the sector were fully formalized and backed by host governments (Figure 9).

The sector's three contributions highlighted here – ASM's contribution to mineral production, impact on employment and the implication of its links with agriculture – are among the more

obvious and significant in sub-Saharan Africa, and which resonate most powerfully with the region's SDG agenda. The next and final section of the article identifies ways in which host governments and donors can go about making a formalized ASM sector more of a centerpiece of SDG strategy in Sub-Saharan Africa, and what must change in order for this to happen.

CHAPTER 6: Conclusion And Recommendations

6.1 Overview

When will sub-Saharan Africa really benefit from an organized ASM sector? The region's extractive industries and development plans are unlikely to evolve beyond supporting foreign-financed, huge resource extraction, which is not usually in the best interests of host nations, unless there is a fundamental transformation in how people see ASM. Mapping Mining to the Sustainable Development Goals: A Preliminary Atlas (UNDP, 2016a), whose content has been criticized throughout this article, confirms the lack of evolution in donor and policymaker perspectives on extractive industries and development, despite the emergence of a dynamic set of targets such as the SDGs. The research demonstrates how large-scale, capital-intensive mining may assist accomplish the Sustainable Development Goals.

In addition, it notes that 'artisanal and small-scale mining is evaluated when it directly connects to LSM' and so avoids analyzing ASM in this context since it "requires a different guide and evaluation to map the opportunities for ASM to contribute to the Sustainable Development Goals." (p. 16). However, the irony of this conscious decision should not be overlooked: the successful management, regulation, and support of ASM is essential for the full realization of the benefits the Atlas connects to large-scale mining in developing countries. Due to the absence of tenure security, the bulk of ASM operators in sub-Saharan Africa have been driven into the unprotected informal sector, where they have come into direct contact with large-scale operators. In locations such as the Democratic Republic of the Congo, Mali, Ghana, Tanzania, and Mozambique, disputes over mining have erupted into violence, giving a context for ASM "to be regarded when it directly relates to LSM" in the Atlas.

Global policy frameworks like the Sustainable Development Goals (SDGs) must identify and react to all branches of particular industrial sectors in resource-poor places like sub-Saharan Africa to maximise economic and social advantages from mining. Despite their different impacts on development, large-scale mining and ASM are both extractives. Specifically, host governments' most "game-changing" macroeconomic contribution is its money. Since each firm's

partnerships with poor nations' governments vary, its advantages to local communities are harder to predict. The Atlas calls the sector's local impact "potential" and stresses the need for good environmental and social standards.

This article examines the most evident instances of this impact in Sub-Saharan Africa. Once these advantages are outlined, it is clear to see how an ASM sector that is legally recognized and well-supported may assist host African nations in accomplishing SDG-driven economic and development objectives. Donor and government circles are becoming more aware of the 'livelihoods' aspect of ASM in rural sub-Saharan Africa, but the sector is still largely absent from regional development policy agendas and discussions around the SDGs. This might be due to a lack of sufficient information upon which to make decisions at this time. A formalization strategy for ASM cannot be implemented, much less designed in a way that speaks directly to the SDGs, given the absence of data on the number of miners, their motivations for mining, the quantity of minerals produced, the location of workers, and the amount of minerals mined.

6.2 Recommendations for future debate

Consequently, the first step is to collect sector-specific data appropriate for mapping onto the SDGs. In 2016, the World Bank and the international NGO PACT worked to fight the dearth of ASM statistics. The groups published DELVE Platform; a platform designed to enable data interchange on ASM. As previously indicated, outdated ASM census and production numbers on sub-Saharan Africa provide a false impression of the sector and have sometimes led to ill-advised government measures. Given the informal character of the sector as a whole, data collecting has been difficult. DELVE is an endeavor to combine the many research on ASM conducted by the government, private enterprises, and universities in the past. Projects in sub-Saharan Africa that aim to gather data on ASM are unlikely to encounter significant pushback, since both the local government and financiers can comprehend this problem.

Several countries in sub-Saharan Africa lack the capacity to collect, manage, and report on demographic, social, economic, and environmental data, which is "crucial for governments to be able to develop better policies and interventions and respond more effectively to national development challenges" as a result of the SDGs' increased emphasis on this area (SDGC, 2018, p. 5). Africa has the lowest average statistical capabilities, and between 1990 and 2012, the average African country conducted just 1.6 comparable demographic surveys (Beegle et al., 2016). Surveys of ASMs might be included into projects aiming to address the SDGs on interconnected subjects such as gender, youth, and food security, therefore expanding the reach and effect of the gathered data. Having DELVE conduct these polls will give legitimacy to the findings. After collecting data, SDG-aware strategies for the formalization and support of ASM may be established and implemented.

Schemes may fall into a framework for development, the most renowned of which is the AMV, whose major objective is "Boosting Artisanal and Small-Scale Mining." In addition, regional development policy frameworks, such as certain PRSPs and the NEPAD Vision, recognize the role of extractive sectors and development, but fail to effectively highlight ASM. Thus, the framework for further formalization of ASM has been established. To ensure the long-term achievement of the SDGs, it is necessary to implement measures that address their basic challenges, such as food security, gender equality, and environmental preservation.

Perhaps now, more than ever, is the time to emphasize the economic importance of ASM in sub-Saharan Africa. Since the United Nations Economic Commission for Africa (UNECA) hosted the international workshop Building Capacity for Environmental Sustainability in Artisanal and Small Scale Mining in Africa (24–26 May 2017), momentum has increased to develop formalization strategies in the region that address the SDGs and closely align with the AMV's objectives.

6.3 Innovative strategies for ASM

As stated in the preceding literature, there is an urgent need for the ASM sector to be recognized within Sub-Saharan Africa's comprehensive regulatory framework. Because of the existing state of the industry, as noted in the results chapter, the bulk of the operator's revenues are spent on

superfluous costs such as, but not limited to, government bureaucratic actors to merely enable them to conduct their everyday operations. If suitable controls are put in place, and the daily produce of the operators is invested wisely, the miners may become a part of the country's total economic growth. The following are some creative strategies that, if appropriately implied, will assist to accomplish the objective.

1. Training workshops and seminars

According to the results chapter, the vast majority of unlicensed miners learn the method of mining from other miners who have more years of expertise. These miners often learn from one another. In most cases, it is the more senior miners that make the first attempts to develop novel production techniques and processes. It would be beneficial for the entirety of the mining industry if experienced miners joined together and shared their expertise on a more widespread basis.

For instance, experienced operators can hold training workshops where they share their knowledge of innovative mining practices and help aspiring mining operators increase their daily yield. In these workshops, the experienced operators can also help the aspiring operators increase their daily output. Additionally, the mine owner has the ability to take on the responsibility of conducting such seminars on a monthly basis, which will bring in a greater number of participants and ultimately result in a better grasp of the entire process.

2. Community engagement

The notion that human psychology may manifest itself in a variety of fascinating ways is a well-established and well accepted reality. It makes an effort to align itself with a larger group of individuals who share traits with itself, and the activities that it does are in some way reliant on the actions that are taken by the community as a whole. ASM operators can take advantage

of this feature, that allows them to gather together in an unstructured manner, exchange their expertise on mining techniques and creative strategies, and learn from one another.

Shifting the focus to the chapter on what was discovered, it is clear that most operators get more than just finances from their daily operations. One of these benefits is that they meet a lot of potential miners and build a network with them. This part can be looked into more and put into place in a more formal way. By sharing information with potential buyers, they can get rid of the problem of responsible sourcing for the most part. This strategy aims to provide an atmosphere conducive to the ASM operating and expanding their operation. Programs for community development might include efforts in infrastructure, education, healthcare, and other social services. Providing access to clean water, sanitation, and power, for instance, may enhance the living circumstances of the community as well as the effectiveness of ASM activities. Improving education may equip ASM and their communities with the necessary information and skills to enhance their operations and boost their production.

In addition, community development projects may include measures that enhance community involvement and empowerment, such as community-based organisations, participatory planning, and deliberative processes. This may aid in fostering confidence and collaboration between the ASM, the community, and local authorities, as well as reducing misunderstandings and disagreements.

3. Implementation and sharing of technology

Building on the strategies mentioned above, ASM operators can pool in their resources in a shared manner in order to purchase/rent machinery. This technique entails using technology to enhance the productivity and viability of artisanal small-scale mining (ASM) activities. One such example is the use of smartphones. Mobile applications can provide small-scale miners with access to a vast array of information and resources that can contribute in the enhancement of their operations. For instance, they may utilize mobile applications to obtain geological data and maps, which can assist them in identifying regions with significant mineral concentrations. In addition, mobile applications may give access to weather predictions and safety

requirements, enabling miners to plan their operations and operate safely. Small-scale miners who may not have access to standard banking services may also utilize mobile money transfer services to sell and exchange minerals more conveniently and securely.

4. **Branding and marketing**

Being part of a large community of unlicensed operators, where the government support is minimal, the ASM industry can take matters into their own hands. One such strategy building on this narrative is branding and marketing. This method entails establishing a brand identity for the artisanal small-scale mining (ASM) operation and advertising it to prospective buyers. By building a brand and publicising it, the ASM company can separate itself from other suppliers and attract to consumers who are prepared to pay a premium for sustainably sourced, high-quality minerals.

Acting as a sub-strategy to point 3 highlighted above, ASM may use numerous marketing platforms, including social media, and offline marketing such as mouth-to-mouth marketing methods, to engage with buyers and promote their business. In other words, miners can act as a small business fostering a sense of community and pooling resources such as finance and knowledge to build this.

6.4 **Conclusion**

Even while the SDGs were not created with ASM in mind, they provide more flexibility than the (Millennium Development Goals) MDGs did. While the lack of specificity in the SDGs has been cited as a major flaw by some commentators (e.g. Murray, 2015; IISD, 2016), we view it as a benefit because it allows us to examine the policies put in place to address larger-scale development challenges and pursue ideas (in this case, those related to the formalization of ASM) that were overlooked during the original design process. Another criticism of the SDGs is that they include too many ambitious objectives (Liverman, 2018). However, it would seem that UN authorities are aware of the magnitude of the work necessary to attain these objectives, and have thus formulated appropriate principles and guidelines and formed international, regional,

and national committees to assist countries. The UN High-Level Political Forum and the UNDP Roadmap for Localizing the SDGs: Implementation and Monitoring at Subnational Level are two essential tools for ASM in sub-Saharan Africa.

One group "is tasked with ensuring the integration of the three dimensions of sustainable development at all levels in a holistic and cross-sectoral manner," while another group "aims to support local and national governments and their associations to implement and monitor the SDGs and to influence national policy making in order to create an enabling environment for action at the local and regional level" (p. 2). 2017 (Stafford-Smith et al.). These panels and guidelines enable the development of new ASM formalization plans and the evaluation of existing ones in sub-Saharan Africa, with the aim of aiding governments in the region in achieving regional SDG-related targets.

This research has shown that formalizing ASM in sub-Saharan Africa would dramatically decrease poverty and enhance economic growth. Three of the sector's most obvious advantages were emphasized, including its capacity to generate revenue, its significance as a source of employment, and its linkages to subsistence agriculture, which all correlate to the SDGs' underlying principles. Although the topics addressed in Mapping Mining to the Sustainable Development Goals: A Preliminary Atlas are significant, a more comprehensive assessment of these and related contributions necessitates the production of a new Atlas (UNDP, 2016a). The consequences of ASM on development in resource-poor regions are significantly different from those of large-scale mining. One of these regions is Sub-Saharan Africa. However, unique policies must be established with the goals of each branch of the mining sector in mind if the consequences of either are to be realized and any mining-led development strategy in the region is to be successful.

APPENDIX

Appendix 1 (Dataset cleansing)

```
# Dataset cleaning
# Removing unnecessary columns from the dataset

newdataframe = subset(dataframe, select = -c(`Date of Survey`, `start`, `end`, `Date of
Survey`, `Part 1 Background`, `Part B: Licensing/Technology General`, `Part 3: Innovation
and Sources of Knowledge`, `Part 4: Networking and Knowledge Flows`, `_id`, `_uuid`,
`_submission_time`, `_validation_status`, `_notes`, `_status`, `_submitted_by`, `_tags`,
`_index`))
dataframe <- newdataframe
view(dataframe)
```

Appendix 2 (Feature Construction)

#Transforming Age into ordinal values

```
dataframe$Age1 <- dataframe$Age

dataframe$Age1 <- str_replace_all(dataframe$Age1, "15-25", "Young Adults")
dataframe$Age1 <- str_replace_all(dataframe$Age1, "26-35", "Early Professionals")
dataframe$Age1 <- str_replace_all(dataframe$Age1, "36-45", "Middle Aged Professionals")
dataframe$Age1 <- str_replace_all(dataframe$Age1, "46-60", "Experienced Professionals")
dataframe$Age1 <- str_replace_all(dataframe$Age1, "60+", "Elderly")
```

#Transforming Number of Years in Mining into categories

```
Experience_in_mining <- dataframe$`Number of years in mining`
Experience_in_mining_num <- as.numeric(Experience_in_mining)
class(Experience_in_mining_num)

df$Experience <- Experience_in_mining_num
```

```
Experience_in_mining <- dataframe$`Number of years in mining`
```

```
df["Experience"] = cut(df$Experience, c(0, 5, 11, 16, 22, 28, 34, Inf), c("0-5", "6-11", "12-16", "17-22", "23-28", "29-34", "35+"), include.lowest=TRUE)
```

Assigning weights to qualitative columns

```
df$Barriers1 <- dataframe$`Barriers for not getting a license`
df$Barriers1 <- str_replace_all(df$Barriers1, "Not applicable", "0")
df$Barriers1 <- str_replace_all(df$Barriers1, "Bureaucracy", "1")
df$Barriers1 <- str_replace_all(df$Barriers1, "Bureaucracy & Others", "2")
df$Barriers1 <- str_replace_all(df$Barriers1, "Bureaucracy & Unaware about the requirements", "3")
df$Barriers1 <- str_replace_all(df$Barriers1, "Costs", "4")
df$Barriers1 <- str_replace_all(df$Barriers1, "Costs & Bureaucracy", "5")
df$Barriers1 <- str_replace_all(df$Barriers1, "Costs & Other", "6")
df$Barriers1 <- str_replace_all(df$Barriers1, "Costs & Unaware about the requirements", "7")
df$Barriers1 <- str_replace_all(df$Barriers1, "Not eligible", "8")
df$Barriers1 <- str_replace_all(df$Barriers1, "Other", "9")
df$Barriers1 <- str_replace_all(df$Barriers1, "Other & Unaware about the requirements", "10")
df$Barriers1 <- str_replace_all(df$Barriers1, "Unaware about the requirements", "11")
```

Appendix 3 (Code for Chi square test, Fisher's exact test & Mosaic Plot)

```
# Mining license v/s Education
```

```
print(chisq.test(Mining_license_vs_Education_table, p = p, rescale.p = TRUE))
```

```
chisq.test(Mining_license_vs_Education_table)$expected
```

```
#Mosaic Plot
```

```
par("mar")
```

```
par(mar=c(1,1,1,1))
```

```

#Plotting the graph
mosaicplot(Mining_license_vs_Education_table,
            main = "Mosaic plot",
            color = TRUE
)

#IFisher's exact test
Mining_license_vs_Education_result <- fisher.test(Mining_license_vs_Education_table)
Mining_license_vs_Education_result

# Mining license v/s Improvements
print(chisq.test(Mining_license_vs_Improvements_table, p = p, rescale.p = TRUE))
chisq.test(Mining_license_vs_Improvements_table)$expected

#Mosaic Plot
#Setting the graph margins
par("mar")
par(mar=c(1,1,1,1))

#Plotting the graph
mosaicplot(Mining_license_vs_Improvements_table,
            main = "Mosaic plot",
            color = TRUE
)

#IFisher's exact test
# Running the fisher test
Mining_license_vs_Improvements_result <-
fisher.test(Mining_license_vs_Improvements_table)
Mining_license_vs_Improvements_result
#Finding the p value in the test
Mining_license_vs_Improvements_result$p.value

fisher_test(Mining_license_vs_Improvements_table, detailed = TRUE, conf.level = 0.95,
conf.int = TRUE)

```

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